Summation Notation or Sigma $\left(\sum\right)$ Notation

$$\sum_{i=m}^{n} x_i$$

i : index of summation

m : lower bound/limit

n : upper bound/limit

x_i : formula or pattern that defines each term in the sum

1) $\sum_{n=1}^{3}(3n+2) = 3(1)+2+3(2)+2+3(3)+2 = \underline{\quad ? \quad}$

2) $\sum_{n=0}^{6}\frac{1}{n^2+1} = \frac{1}{(0)^2+1}+\frac{1}{(1)^2+1}+\ldots+\frac{1}{(6)^2+1} = \underline{\quad ? \quad}$

3) $\sum_{n=1}^{4}5 = 5+5+5+5 \ = \underline{\quad ? \quad}$

4) $\sum_{n=1}^{10}7 = 7+7+7+\ldots+7 = \underline{\quad ? \quad}$

5) a) $\sum_{i=1}^{4} 3i = \underline{\quad ? \quad}$

b) $3 \cdot \left(\sum_{i=1}^{4} i \right) = \underline{\quad ? \quad}$

6) a) $\sum_{i=1}^{4} 5i^2 = \underline{\quad ? \quad}$

b) $5 \cdot \left(\sum_{i=1}^{4} i^2 \right) = \underline{\quad ? \quad}$

7) a) $\sum_{i=1}^{4} 6i^3 = \underline{\quad ? \quad}$

b) $6 \cdot \left(\sum_{i=1}^{4} i^3 \right) = \underline{\quad ? \quad}$

8) a) $\sum_{i=1}^{3} \left(2i + i^2 \right) = \underline{\quad ? \quad}$

b) $\sum_{i=1}^{3} (2i) + \sum_{i=1}^{3} \left(i^2 \right) = \underline{\quad ? \quad}$

9) a) $\sum_{i=1}^{3} \left(4i - i^2 \right) = \underline{\quad ? \quad}$

b) $\sum_{i=1}^{3} (4i) - \sum_{i=1}^{3} \left(i^2 \right) = \underline{\quad ? \quad}$

In general: $\sum_{i=1}^{n} c \cdot i = c \cdot \sum_{i=1}^{n} i;$ $\qquad \sum_{i=1}^{n} (A \pm B) = \sum_{i=1}^{n} (A) \pm \sum_{i=1}^{n} (B)$

Summation Formulas

$$\sum_{i=1}^{n} c = c \cdot n \qquad c \text{ is a constant}$$

$$\sum_{i=1}^{n} i = 1 + 2 + 3 + \ldots + n = \frac{n^2}{2} + \frac{n}{2}$$

$$\sum_{i=1}^{n} i^2 = 1^2 + 2^2 + 3^2 + \ldots + n^2 = \frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6}$$

$$\sum_{i=1}^{n} i^3 = 1^3 + 2^3 + 3^3 + \ldots + n^3 = \frac{1}{4}n^4 + \frac{1}{2}n^3 + \frac{1}{4}n^2$$

$$\sum_{i=1}^{n} i^4 = 1^4 + 2^4 + 3^4 + \ldots + n^4 = \frac{1}{5}n^5 + \frac{1}{2}n^4 + \frac{1}{3}n^3 - \frac{1}{30}n$$

10) $\sum_{n=1}^{50} i = 1 + 2 + 3 + \ldots + 50 = \underline{\quad ? \quad}$

12) $\sum_{n=1}^{50} i^2 = 1^2 + 2^2 + 3^2 + \ldots + 50^2 = \underline{\quad ? \quad}$

13) $\sum_{n=1}^{50} i^3 = 1^3 + 2^3 + 3^3 + \ldots + 50^3 = \underline{\quad ? \quad}$

14) $\sum_{i=1}^{n} 3 \cdot i = 3 \cdot \sum_{i=1}^{n} i = \underline{\quad ? \quad}$

15) $\sum_{i=1}^{n} 5n^2 \cdot i = 5n \cdot \sum_{i=1}^{n} i^2 = \underline{\quad ? \quad}$

16) $\sum_{i=1}^{n} 7n^2 \cdot i^3 = 7n^2 \cdot \sum_{i=1}^{n} i^3 = \underline{\quad ? \quad}$

17) Write $\sum_{i=1}^{n}\left(2i+i^2\right)$ in terms on n only.

$$\sum_{i=1}^{n}\left(2i+i^2\right) = \sum_{i=1}^{n}(2i) + \sum_{i=1}^{n}\left(i^2\right) = 2 \cdot \sum_{i=1}^{n}(i) + \sum_{i=1}^{n}\left(i^2\right) = \underline{\quad ? \quad}$$

18) Write $\sum_{i=1}^{n}\left(\frac{3}{n} \cdot i - \frac{2}{n^2} \cdot i^2\right)$ in terms on n only.

$$\sum_{i=1}^{n}\left(\frac{3}{n} \cdot i - \frac{2}{n^2} \cdot i^2\right) = \sum_{i=1}^{n}\left(\frac{3}{n} \cdot i\right) - \sum_{i=1}^{n}\left(\frac{2}{n^2} \cdot i^2\right) = \frac{3}{n} \cdot \sum_{i=1}^{n}(i) - \frac{2}{n^2} \cdot \sum_{i=1}^{n}\left(i^2\right) = \underline{\quad ? \quad}$$

19) Write $\sum_{i=1}^{n}\left(\frac{1}{n^2}\cdot i^3-\frac{5}{n^2}\cdot i^2\right)$ in terms on n only.

$$\sum_{i=1}^{n}\left(\frac{1}{n^2}\cdot i^3-\frac{5}{n^2}\cdot i^2\right)=\sum_{i=1}^{n}\left(\frac{1}{n^2}\cdot i^3\right)-\sum_{i=1}^{n}\left(\frac{5}{n^2}\cdot i^2\right)=\frac{1}{n^2}\cdot\sum_{i=1}^{n}\left(i^3\right)-\frac{5}{n^2}\cdot\sum_{i=1}^{n}\left(i^2\right)=\underline{\quad ?\quad}$$

20) $\sum_{i=1}^{n}\frac{15}{n^2}=\underline{\quad ?\quad}$

Note: Summation is with respect to i;

The expression $\frac{15}{n^2}$ does not have i; $\frac{15}{n^2}$ is considered constant